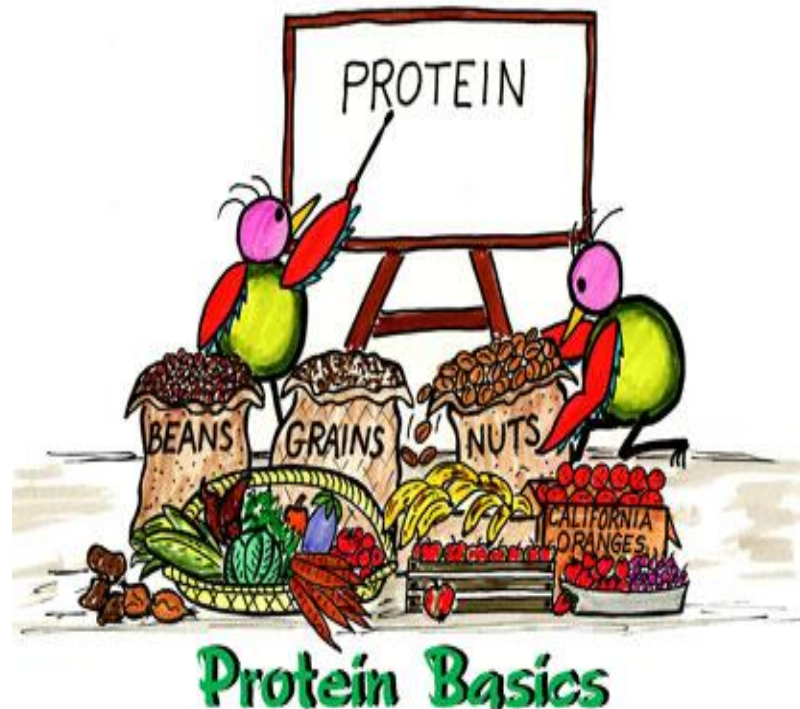


CLASSIFICATION & FUNCTIONS OF PROTEINS



DR.HUMAIRA AMAN

PROTEINS

- Greek word : protos means first or supreme.
- DEFINITION :

A molecule made up of amino acids that are needed for the body to function properly .

OR

Proteins are linear polymers of amino acids.

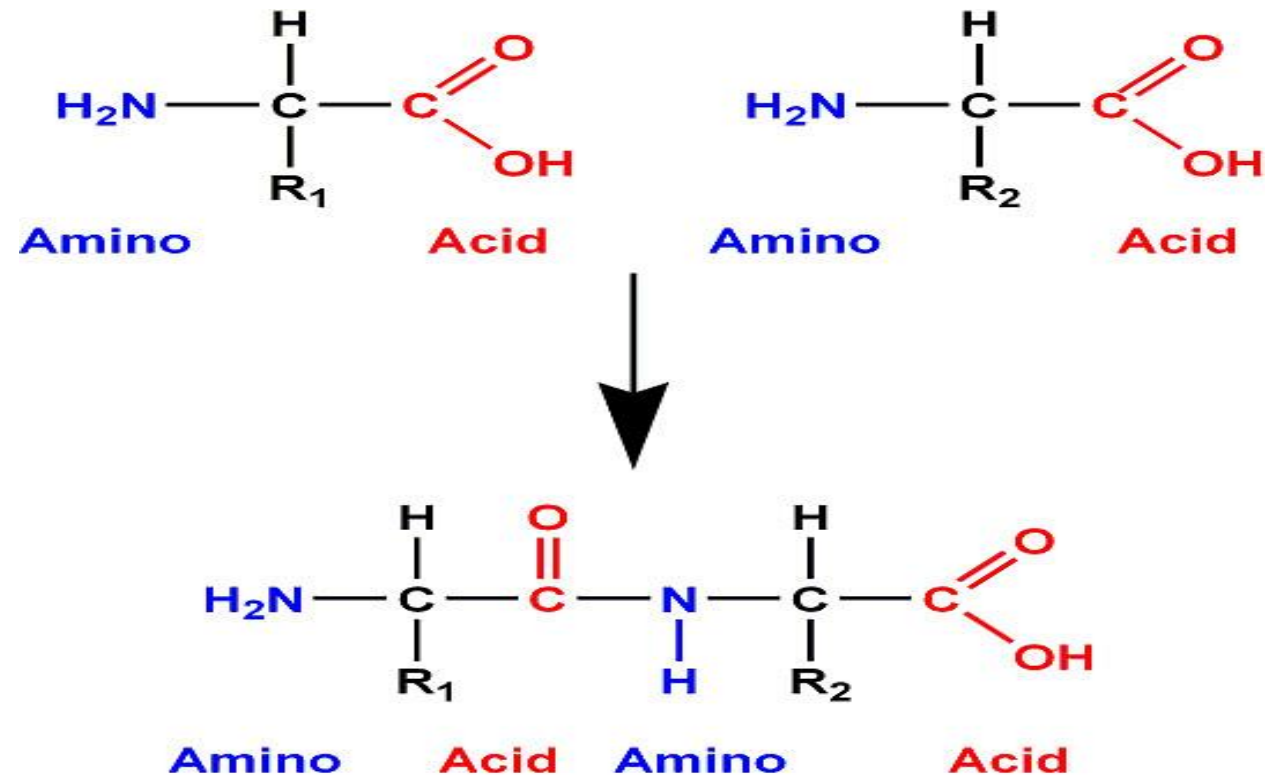
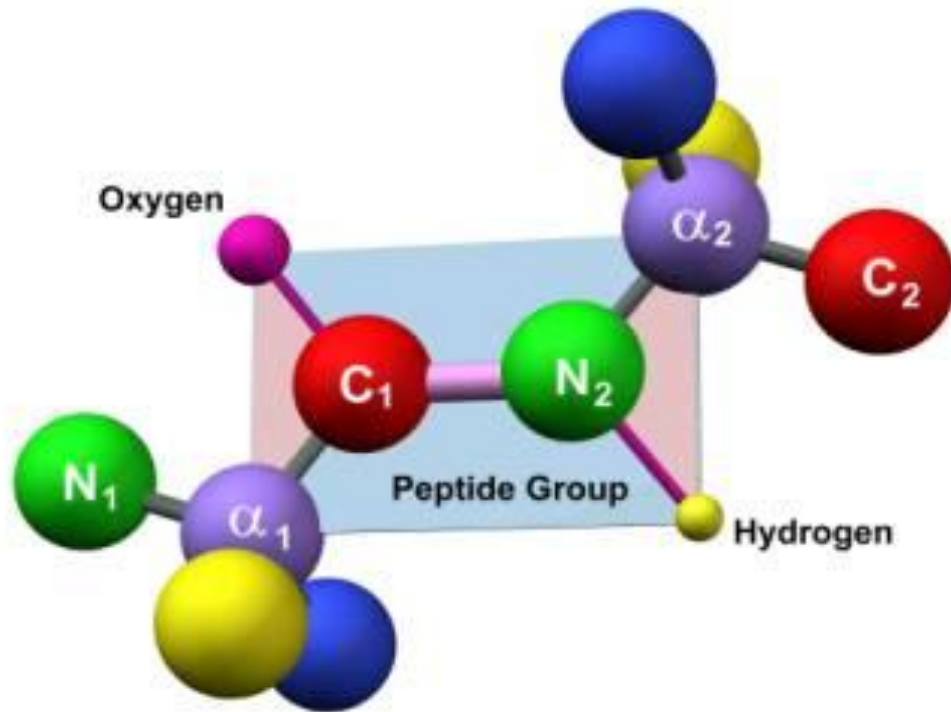
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- Any of a group of complex organic macromolecule that contains carbon, hydrogen , oxygen , nitrogen and usually sulfur and are composed of one or more chains of amino acids.
- Proteins are extremely complicated molecule and are nitrogenous compounds made up of a variable number of amino acid residues joined to each other by peptide linkages.

- They are the most abundant and functionally diverse molecules in living system .
- They are the most abundant organic molecules in the cell interior where they constitute 50% or more of their dry mass .
- Virtually every life process depends on this class of molecules in livings.
- Eg :
 - ❖ Enzymes and hormones regulate metabolism.
 - ❖ Contractile proteins in muscles permit movement.
 - ❖ In bones , collagen protein forms frame work .
 - ❖ In blood, Hb , albumin , Immunoglobulins.

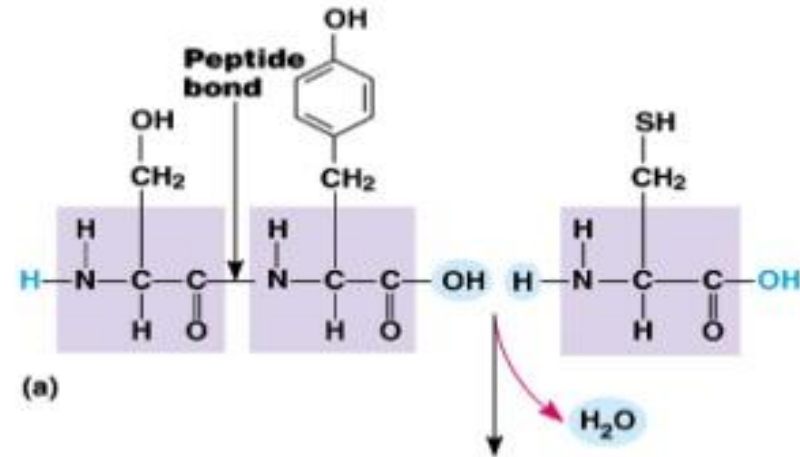
THE PEPTIDE LINKAGE OR PEPTIDE BOND

- The amino acids are coupled by a PEPTIDE BOND, a special linkage in which the nitrogen atom of one amino acid binds to the carboxyl carbon atom of neighbouring amino acid thus formed ACID-AMIDE BOND called a PEPTIDE BOND

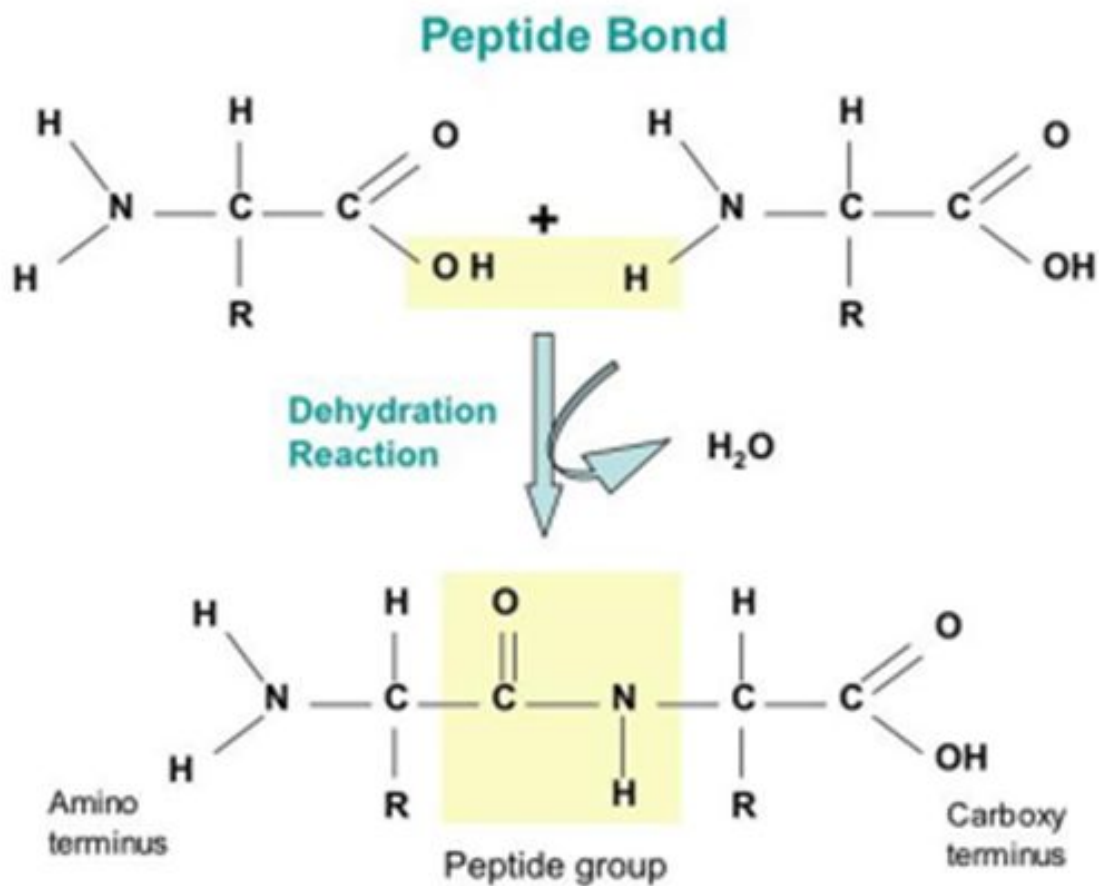


Forming a Polypeptide

- Condensation reaction to join 2 amino acid
- Requires:
 - Carboxyl group
 - Amine



Peptide Bonds



PEPTIDES & PROTEINS

- Peptide is a molecule formed by joining two or more amino acids .
- The chains containing less than 50 amino acids are called “peptides”, while those containing greater than 50 amino acids are referred to as PROTEINS (POLYPEPTIDES) .
- Polypeptides range in length from a few to more than a thousand monomers.
- Each polypeptide has a unique linear sequence of amino acids, with a carboxyl end (C-terminus) and an amino end (N-terminus).

Protein shape determines function

- Amino acid order determines the shape (conformation) of a protein
- Conformation determines function
- Function depends on its ability to recognize and bind a molecule

Amino acids → conformation → function → binding

CLASSIFICATION OF PROTEINS

- Proteins have been classified on the following ways based on their function, chemical nature and solubility and nutritional importance.
1. *Chemical nature and solubility* (PHYSICOCHEMICAL)
 2. *Functional classification of proteins.*
 3. *Nutritional importance.*

1. **CLASSIFICATION BASED UPON CHEMICAL PROPERTIES OF PROTEINS** :

Proteins may be classified into three types.

- ✓ Simple proteins
- ✓ Compound or conjugated proteins
- ✓ Derived proteins

Proteins

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graph TD; A[Proteins] --> B[Simple Proteins]; A --> C[Conjugated Proteins]; A --> D[Derived proteins];
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Simple Proteins

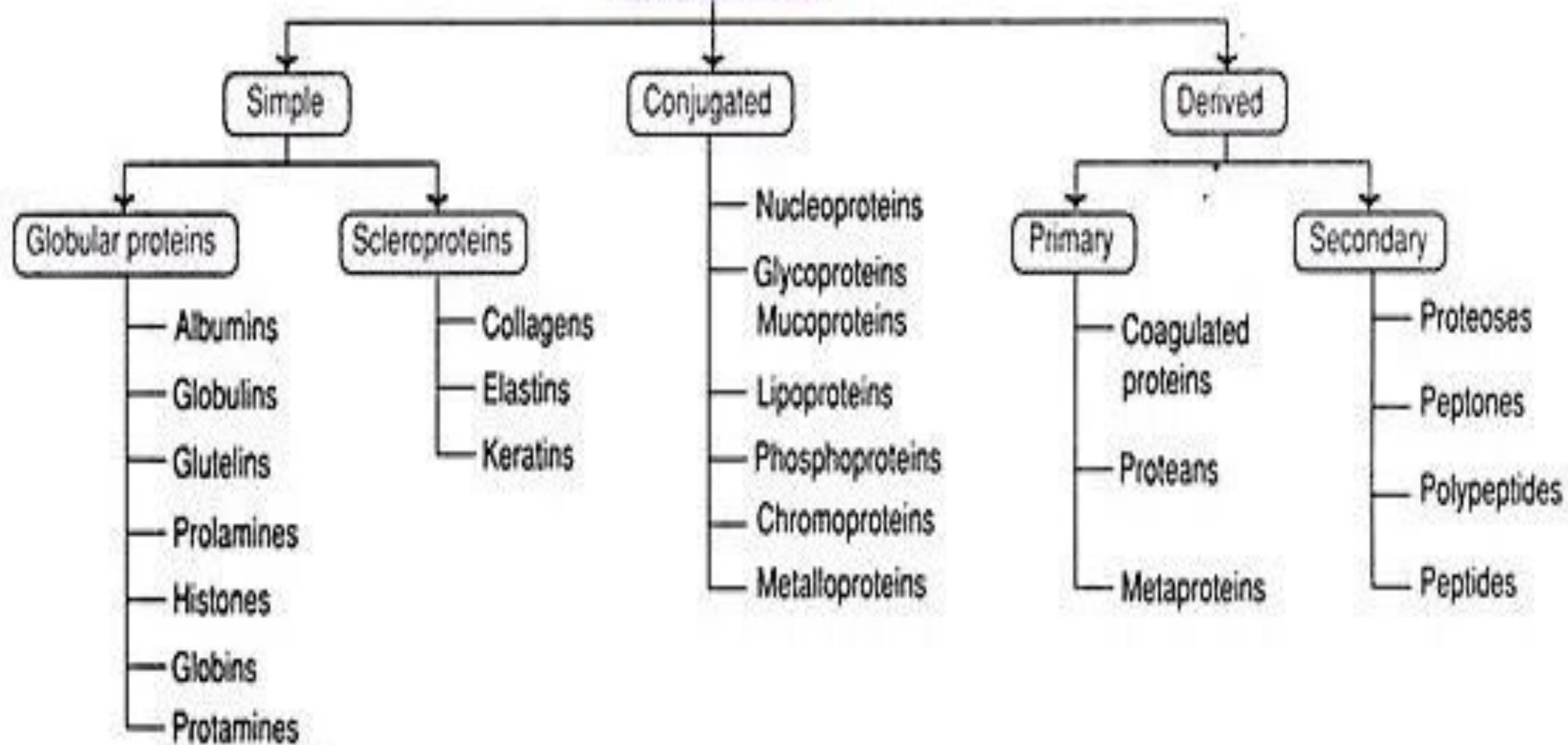
- Protamine
- Histone
- Albumin
- Globulin
- Epidermal

Conjugated Proteins

- Nucleoproteins
- Chromoproteins
- Phosphoproteins
- Glycoproteins and mucoproteins
- Lipoproteins
- Metalloproteins

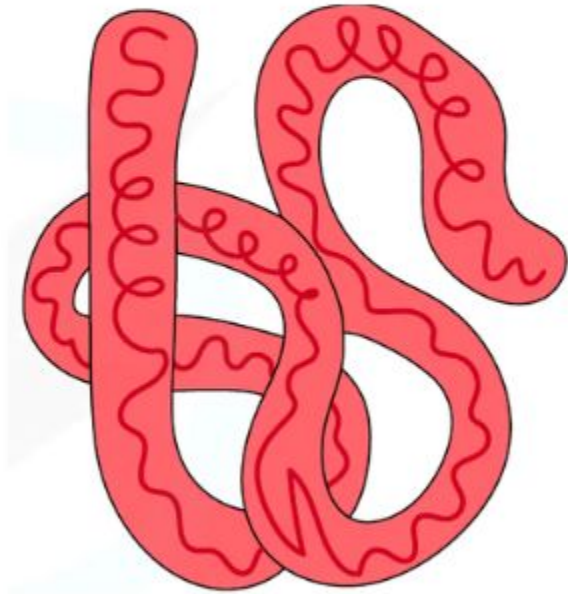
Derived proteins

PROTEINS



Globular Protein

Globular proteins are spherical and function in transport, enzymes, and regulation



Vs

Fibrous Protein

Fibrous proteins are elongated and structural, providing support and strength to tissues.



SIMPLE PROTEINS

“On hydrolysis (addition of water molecule at the site of peptide bond causing breakdown of protein molecules giving rise to amino acids), these proteins yield **only amino acids**.”

A) Globular proteins:

- These are spherical or oval in shape, soluble in water or other solvents and digestible.

For example:

- **Albumins:**

- ☐ They are soluble in water and coagulated by heat. E.g serum albumin, ovalbumin (egg), lactalbumin (milk).

- **Globulins:**

- ☐ These are insoluble in pure water, but soluble in dilute salt solutions and coagulated by heat.
- ☐ They are precipitated by half saturation with ammonium sulphate or by full saturation with sodium chloride. E.g., serum globulins, egg globulins.

- **Glutelins:**

- ☐ These are plant proteins, insoluble in water or neutral salt solutions, soluble in dilute acids or alkalies.
- ☐ They are rich in **glutamic acid**.
- ☐ They are large molecules and can be coagulated by heat. e.g., glutelin (wheat), oryzenin (rice).

- **Prolamines:**

- ❑ They are soluble in 70-80% alcohol, but insoluble in pure water.
- ❑ They are rich in **proline** but lack in lysine. E.g., gliadin (wheat) and zein (maize).

- **Histones:**

- ❑ These are basic proteins, rich in **arginine and histidine**.
- ❑ In combination with **DNA and RNA** they form nucleoproteins {conjugated proteins).
- ❑ They are soluble in water, dilute acids and salt solutions but insoluble in ammonia.

- **Globins:**

- ☐ Globin refers to a family of heme-containing globular proteins that are in charge of binding and transporting oxygen.

- **Protamines:**

- ☐ Protamines are small, arginine-rich nuclear proteins found in the sperm of most vertebrates, including humans, mammals, and fish (e.g., salmon).
They replace histones during spermatogenesis to compact DNA, crucial for sperm head stabilization.
- ☐ These are soluble in water, dilute acids and alkalies. They are not coagulated by heating..

B: Scleroproteins or Albuminoids

These are fibrous proteins with great stability and very low solubility and form supporting structures.

- **Collagen:**

- ☐ Collagen is the connective tissue protein, lacking tryptophan and high in glycine, proline, and hydroxyproline.
- ☐ Most abundant protein in human body, forming approximately 25% to 30% of body proteins.

cont..

- **Elastin:**

- ☐ This is also a connective tissue protein occur in elastic tissues eg: tendons and arteries .

- **Keratin:**

- ☐ These are present in exoskeletal structures(ectodermal tissues) e.g. hairs, nails, horns.

* Human hair has a higher content of cysteine.

COMPOUND OR CONJUGATED PROTEINS:

In these molecules the protein is attached or conjugated to some non-protein groups which are called as PROSTHETIC GROUPS. e.g.

- **NUCLEOPROTEINS :**

- ☐ Simple protein “histone” conjugates with nucleic acid (DNA and RNA).

- **PHOSPHOPROTEINS:**

- ☐ Proteins conjugated with phosphoric acid .eg: casein in milk and vitelline of egg yolk.
- ☐ The phosphoric acid is esterified to the hydroxyl groups of serine and threonine residues of proteins.

cont

- **CARBOHYDRATE-CONTAINING PROTEINS or GLYCOPROTEINS:**

- ☐ Glycoproteins are the proteins with carbohydrate moiety as the prosthetic group.
- ☐ **Mucoproteins** : having more than 4% carbohydrates .

- **LIPOPROTEINS :**

- ☐ These are the proteins loosely combined with lipid components.
- ☐ They occur in blood and on cell membranes.
- ☐ Serum lipoproteins, membrane lipoproteins

cont

- **CHROMOPROTEINS:** These are the proteins with colored prosthetic groups.
 - **Hemoproteins:** All hemoproteins are Chromoproteins which carry heme as the prosthetic group.e.g. Hemoglobin
 - **Cytochromes:** These are the mitochondrial enzymes of the respiratory chain.
- **METALLOPROTEINS:** proteins with metallic atoms . Eg: Ferritin (Fe) , carbonic anhydrase (Zn) and ceruloplasmin (Cu).

DERIVED PROTEINS:

This class of proteins includes substances which are derived from simple and conjugated proteins .

These are subdivided into primary and secondary derived proteins.

PRIMARY DERIVED PROTEINS:

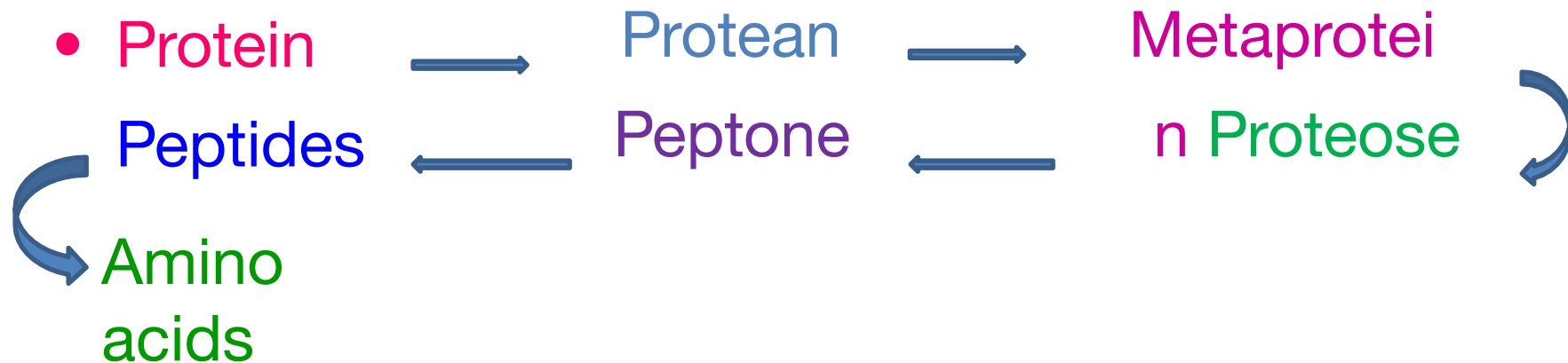
- ☐ Synonym: Denatured proteins.
- ☐ A protein is called native protein if its amino acids composition and molecular conformation are unchanged from that found in the natural states .These properties control all the functions of a protein .
- ☐ Denaturation takes place when some or all of the cross linkages which normally keep the molecular structure of a protein intact are split .Although there is no or partial hydrolysis takes place resulting in the loss of certain amino acids or peptides.
- ☐ Denaturation may be brought about by many chemical or physical agents .

- **Coagulated proteins:** resulting from the action of heat and alcohol are classed in this division.
- **Proteans:** These are earliest products of protein hydrolysis by enzymes, dilute acids, alkalies etc. They are insoluble in water
 - E.g., Myosan (from myosin), Edestan (from elastin)
- **Meta proteins:** These are the second stage products of protein hydrolysis by treatment with slightly stronger acids and alkalies.

SECONDARY DERIVED PROTEINS

- These substances are intermediates formed in the progressive hydrolysis of protein molecules.
- They are of different sizes and are roughly grouped into proteoses and peptones.
- **PROTEOSE** is an intermediate stage between the original protein and a peptone, and it receives a name according to its source, as albumose, globulose, and caseose, according as an albumin, a globulin, or casein is its source.
- **PEPTONES** are of simpler structure than proteoses further hydrolyzed into polypeptides which in turn give rise to fragments of still shorter chain lengths called oligopeptides.

- **Hydrolysis:** The complete hydrolytic decomposition of a protein generally follows these stages



2: Functional classification of proteins

- ❑ Structural proteins: Keratin of hair and nails, collagen of bone
- ❑ Enzymes or catalytic proteins: Hexokinase, pepsin
- ❑ Transport proteins: Hemoglobin, serum albumin
- ❑ Hormonal proteins: Insulin, growth hormone

cont..

- ❑ Contractile proteins: Actin, myosin
- ❑ Storage proteins: Ovalbumin, glutelin
- ❑ Genetic proteins: Nucleoproteins
- ❑ Defense proteins: Immunoglobulins
- ❑ Receptor proteins: For hormones and viruses

3. Nutritional classification of proteins

- Complete proteins or Nutritionally rich proteins:
 - ❑ These proteins have all the essential amino acids in the required proportion.
 - ❑ Also called as first class proteins. E.g., egg albumin, casein of milk
- Partially incomplete proteins:
 - ❑ These proteins are partially lacking one or more essential amino acids and can promote moderate growth. E.g., wheat and rice proteins.
 - ❑ Limiting amino acids Lysine and Threonine.

- Incomplete or poor proteins:
- They lack in many essential amino acids Hence they do not promote growth at all. E.g. processed meat and refined grains.

Biologically Important Peptides

- Biologically active peptides:

1. Glutathione:

- It is tripeptide composed of 3 amino acids.
- Chemically, glutathione is γ -glutamyl-cystinyl- glycine.
- It exists in reduced or oxidized states.

Functions of Glutathione:

- Glutathione maintains RBC membrane structure and integrity.
- Glutathione protects hemoglobin from getting oxidized by agents such as H_2O_2 .
- Glutathione is involved in the transport of amino acids in the intestine and kidney tubules via γ -glutamyl cycle.
- Glutathione involved in the detoxification process.
- Toxic amounts of peroxides and free radicals produced in the cells are scavenged by glutathione peroxidase (a selenium containing enzyme)

2. Thyrotrophin releasing hormone(TRH):

- It is tripeptide secreted by hypothalamus.
- TRH stimulates the release of TSH.

3. Oxytocin:

- It is a hormone secreted by posterior pituitary gland and contains 9 amino acids (nonapeptide).
- Oxytocin causes contraction of uterus

4. Vasopressin (antidiuretic hormone,ADH):

- ADH is also a nonapeptide secreted by posterior pituitary gland.
- It stimulates kidney to retain water and thus increases blood pressure.

5. Angiotensins:

- Angiotensin I is a decapeptide (10 amino acids) which is converted to angiotensin II (8 amino acids)
- Angiotensin II stimulates the release of aldosterone from adrenal gland

**Thank
you**